Emily Fairfield

Data Scientist

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EDUCATION

MASTER OF SCIENCE, COMPUTER SCIENCE, ML CONCENTRATION | Georgia Institute of Technology | GPA: 3.8/4.0 Dec. 2023
MASTER OF BUSINESS ADMINISTRATION (MBA) | Kennesaw State University | GPA: 4.0/4.0 May 2020
BACHELOR OF SCIENCE, INDUSTRIAL & SYSTEMS ENGINEERING | Georgia Institute of Technology | Magna cum laude Dec. 2015

WORK EXPERIENCE

US AIR FORCE CIVILIAN SERVICE 2017 - Present

LEAD DATA SCIENTIST, Special Operations Forces (SOF) US Air Force Civilian Service | Robins Air Force Base (AFB), GA | 2023 - Present

- Aircraft Telemetry Data Science Lead – Lead 2-year-old, first-of-its-kind, government-owned aircraft telemetry data science program for Special Operations aircraft. Manage entire data/ML pipeline.
- Multithreaded, Vectorized Aircraft Telemetry Translator – Developed, tested, and documented multithreaded, vectorized data translator that converts raw byte streams into chronologically-sorted, human-readable data for real-time aircraft health monitoring and intrusion detection on Special Operations aircraft. (Petabytes of data annually). Compared to legacy translator, simultaneously improved runtime by 10x, quantity of data translated by 6x, and output file size by 13x. Enabled real-time aircraft telemetry monitoring on a laptop with remaining laptop resources sufficient for ML model evaluation.
- Term Frequency-Inverse Document Frequency (TF-IDF) and K-Means Clustering - Applied TF-IDF & K-Means Clustering to maintenance records to find prominent issues by aircraft delivery date for contractor quality standards accountability.
- Statistical Analyses - Applied Kolmogorov-Smirnov test to find time ranges of steady level flight for fuel efficiency project.
- Hardware Integration – when unforeseen issues led to test event data being recorded with different hardware version, made software adjustments to ensure data was still useable, avoiding costly duplicate test flight event.
- Collaborated with firmware developers to measure new electrical signal characteristics for intrusion detection based on latest research in electronics fingerprinting.
- Established documentation, peer review, and software testing standards for team. Run weekly sprint planning meetings to prioritize workload according to customer needs, accommodating urgent troubleshooting of aircraft failures.
- Lead collaborative meetings with users – aircraft maintenance & operations personnel – to translate their needs into features, including data visualizations. Collaborate with system engineers to confirm understanding of data.
- Assess & validate contractor-developed AI applications & train engineers of other disciplines to generate AI requirements.

DATA SCIENTIST, Automatic Test Systems (ATS) US Air Force Civilian Service | Robins AFB, GA | 2019 - 2023

- Led first-ever Digital Innovation Team, transforming ambiguous goals into value for internal and external customers. Identified opportunities, developed technical requirements, executed projects, & trained users.
- XML Data Translator - Developed prototype script to clean & transform avionics troubleshooting test result files for customers' ML models. Provides high-fidelity ground truth labels for hundreds of aircraft system failures, compared to prior human labels.
- Nuclear Certification Database – Developed database with MS Access, VBA, & SQL. Incorporated feedback. Wrote manual.
- Cameo Systems Models - Adapted models for ATS configuration management & requirements traceability/management. Trained hundreds of users. Created promotional video.

PROCESS ENGINEER, F-15 Maintenance US Air Force Civilian Service | Robins AFB, GA | 2017 - 2019

- Increased F-15 Aircraft Availability (AA) by 660 days/year and brought $30M/year of workload to Robins Air Force Base by increasing capacity by 150% without building a new facility. Collaborated extensively with cross-functional team of engineers, aircraft maintainers, facility managers, and contractors. Regularly presented project status and recommendations to leadership. Led regular process flow mappings with stakeholders.

MANAGEMENT ENGINEER Northside Hospital | Atlanta, GA | 2016 - 2017

- Labor & Delivery (L&D) Room Turnover - Reduced 70% of L&D patients' stay by 21 min, with cascading benefit to waiting room.
- Managed productivity models for >100 departments, including surgical, postpartum, radiology, and behavioral health.
- Served as consultant to hospital managers, training them to use productivity software & collaborating on process improvements.

PROFESSIONAL AWARDS

2024 General James Ferguson Engineering Award Finalist | 2023 Air Force Civilian Achievement Award | 2022 Junior Engineer of the Year

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MACHINE LEARNING & OPTIMIZATION PROJECTS

Personal Projects – Sample of ML (PyTorch), Data Cleansing/ETL, & Optimization Projects: https://emilyfairfield.github.io/ 2025

Make Data Count – Finding Data References Kaggle Competition Ongoing

- Developing end-to-end ML solution for two-part task: 1) Reading XML & PDF scientific articles, extracting dataset IDs from them, and 2) classifying each article/dataset pair as primary (generated by/for this study) or secondary (reused from another study).
- Wrote regex to capture dataset ID formats from training data, careful to avoid data leakage from validation set. Labeled regex matches & trained a Named Entity Recognition (NER) model using Hugging Face transformer library & SciBERT tokenizer.
- Achieved 97% validation set recall for part 1 by ensembling regex & NER results when generating dataset ID candidates.
- For part 2, performed feature engineering, writing rules to extract features such as: surrounding sentence & paragraph, location of sentence in article as a percentage of article length, and presence of certain “hint” words in surrounding sentence that might help classify it as primary (e.g., “generated”) or secondary (e.g., “downloaded”).
- Applied Principal Component Analysis to embeddings. Scaled numeric features. Used sklearn SelectKBest for feature selection.
- Currently tuning hyperparameters and architectures

Image Classification Explainability Project 2023

Developed and explained deep vision classification models using saliency maps, class visualizations, and Grad-CAM to make CNN behavior transparent and interpretable for non-technical reviewers. Produced image-heavy reports overlaying attention heatmaps onto original images to communicate which pixels in an image had the most influence on a classification decision.

CNN Image Style Transfer with Custom Loss Function 2023

Implemented neural style transfer (Gatys et al., CVPR 2015) to generate hybrid content–style images using a multi-term loss function balancing content and style objectives, demonstrating advanced image-gradient optimization techniques.

Enron Email Corpus Project 2023

Built an interpretable NLP search pipeline combining ElasticSearch, TF-IDF vectorization, and clustering to surface ethically questionable communications from millions of corporate emails, demonstrating how transparent text-mining methods can enhance trust and auditability in complex data systems.

Stock Trading Agent 2022

Developed a Q-learning stock trading agent, achieving out-of-sample returns of 1.23 – higher than manual and benchmark strategies - using five technical indicators and an optimizer for lookback tuning.

NeurIPS 2025 - Google Code Golf Championship Ongoing

- Ongoing Kaggle competition to solve graphical transformation problems using fewest Python characters
- Prompt engineering with Claude Code & ChatGPT
- Developed script for team to check correctness and size of new solutions vs. old ones, trying various compression techniques.
- See our current position on the leaderboard under team name “Emily Anna Chris Brandon” at https://www.kaggle.com/competitions/google-code-golf-2025/leaderboard

SMART on FHIR App – Captured health data & exported to FHIR-compliant format for interoperability with simulated EHR system. 2022

Logistics Decision Support System – Georgia Tech Senior Design (Consultants for Gypsum Management & Supply) 2015

Reduced delivery costs by 6.5% by creating a Python-based logistics tool to load & route trucks while allowing last-minute orders.

SKILLS

Machine Learning | Data Visualization (D3, Matplotlib) | Extract, Transform, Load (ETL) | Project Management | Unit Testing | Strategic Planning | Process Improvement | Technical Requirements Definition | Code Documentation |Python (SciKit Learn, Dask, Pandas, Numpy) | MLOps (MLflow, DVC, DagsHub, Git) | SQL | VBA | Process Flow Mapping | Natural Language Processing

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